

Arne Van Antwerpen

email: arne.vanantwerpen@mu.ie
URL: <https://vanantwerpen.github.io/>

Nationality: Belgian

Current position

2025-current *Assistant Professor/Lecturer*, Department of Mathematics and Statistics, Maynooth University, Ireland

Previous positions held

2023-2025 Post-doctoral assistant in the group of Prof. T. De Medts at Universiteit Gent
2020-2023 FWO post doctoral fellow (junior) in the group of Prof. L. Vendramin at Vrije Universiteit Brussel
2016-2020 FWO phd fellow in the group of Prof. E. Jespers at Vrije Universiteit Brussel

Education

2020 PhD in Science (Mathematics), Vrije Universiteit Brussel
2016 MSc in Mathematics, Universiteit Antwerpen
2014 BSc in Mathematics, Universiteit Antwerpen

Fellowships and awards

COMPETITIVE FELLOWSHIPS

2023 FWO senior post-doctoral fellow - Fonds Wetenschappelijk Onderzoek Vlaanderen, Belgium, Suspended.
2020-2023 FWO junior post-doctoral fellow - Fonds Wetenschappelijk Onderzoek Vlaanderen, Belgium
2016-2020 FWO PhD fellow - Fonds Wetenschappelijk Onderzoek Vlaanderen, Belgium

TRAVEL GRANT

2024 Travel expenses grant of FWO for attending the workshop “Braids, skew braces and the Yang-Baxter equation” at Banff International Research Station, Canada.

CONFERENCE FUNDING

2022 Conference funding FWO and Compositio Mathematica for conference “Groups, rings and Yang-Baxter equation” (joint w. I. Colazzo and L. Vendramin) - 10000 euro

PHD GRANTS

2021-2025 Grant for S.Trappeniers - FWO Phd fellow - Fonds Wetenschappelijk Onderzoek Vlaanderen

Publications & talks

My 5 most important publications are highlighted by ★.

JOURNAL ARTICLES

- 2026 1. Colazzo Ilaria, Van Antwerpen Arne, “On the cabling of non-involutive solutions of the Yang–Baxter equation”, *Journal of Algebra*, published online.
- 2025 2. Properzi Silvia, Van Antwerpen Arne (2025), “A common divisor graph for skew braces”, *Journal of Pure and Applied Algebra*, 229 (2): 107876.
- 2023 3.★ Colazzo Ilaria, Jespers Eric, Kubat Łukasz, Van Antwerpen Arne, Verwimp Charlotte (2023), “Finite idempotent set-theoretic solutions of the Yang–Baxter equation”, *International Mathematics Research Notices*, doi: 10.1093/imrn/rnad183
- 2023 4.★ Jespers Eric, Van Antwerpen Arne, Vendramin Leandro (2023), “Nilpotency of skew braces and multipermutation solutions of the Yang–Baxter equation”, *Communications in Contemporary Mathematics* 25 (09), 2250064.
- 2023 5. Cedó Ferran, Jespers Eric, Kubat Łukasz, Van Antwerpen Arne, Verwimp Charlotte (2023), “On various types of nilpotency of the structure monoid and group of a set-theoretic solution of the Yang–Baxter equation”, *Journal of Pure and Applied Algebra* 227 (2): 107194
- 2022 6. Colazzo Ilaria, Jespers Eric, Van Antwerpen Arne, Verwimp Charlotte (2021), “Left non-degenerate set-theoretic solutions of the Yang–Baxter Equation and semitrusses”, *Journal of Algebra* 610: 409–462
- 2021 7.★ Jespers Eric, Kubat Łukasz, Van Antwerpen Arne, Vendramin Leandro (2021), “Radical and weight of skew braces and their applications to structure groups of solutions of the Yang–Baxter equation”, *Advances in Mathematics* 385: 107767
- 2020 8. Jespers Eric, Kubat Łukasz, Van Antwerpen Arne (2020), “Corrigendum et addendum to The structure monoid and algebra of a non-degenerate set-theoretic solution of the Yang–Baxter equation”, *Transactions of the American Mathematical Society* 373 (6): 4517–4521
- 2019 9. Colazzo Ilaria, Van Antwerpen Arne (2019), “The algebraic structure of left semi-trusses”, *Journal of Pure and Applied Algebra* 225 (2): 106467
- 2019 10.★ Jespers Eric, Kubat Łukasz, Van Antwerpen Arne, Vendramin Leandro (2019), “Factorizations of skew braces”, *Mathematische Annalen* 375 (3–4): 1649–1663
- 2019 11.★ Jespers Eric, Kubat Łukasz, Van Antwerpen Arne (2019), “The structure monoid and algebra of a non-degenerate set-theoretic solution of the Yang–Baxter equation”, *Transactions of the American Mathematical Society* 372 (10): 7191–7223
- 2019 12. Jespers Eric, Van Antwerpen Arne (2019), “Left semi-braces and solutions of the Yang–Baxter equation”, *Forum Mathematicum* 31 (1): 241–263
- 2018 13. Van Antwerpen Arne (2018), “Coleman Automorphisms of finite groups and their minimal normal subgroups”, *Journal of Pure and Applied Algebra* 222 (11): 3379–3394

PRE-PRINT

- 2026 Cascella Rosa, Properzi Silvia, Van Antwerpen Arne, “Finiteness conditions on skew braces and solutions of the Yang–Baxter equation”, ArXiv:2603.06177.
- 2024 Colazzo Ilaria, Okniński Jan, Van Antwerpen Arne, “Bijective solutions to the Pentagon Equation”, ArXiv:2405.20406.
- 2023 Colazzo Ilaria, Jespers Eric, Kubat Łukasz, Van Antwerpen Arne, “Simple solutions of the Yang–Baxter equation”, ArXiv:2312.09687.
- 2023 Colazzo Ilaria, Jespers Eric, Kubat Łukasz, Van Antwerpen Arne, “Structure algebras of finite set-theoretic solutions of the Yang–Baxter equation”, ArXiv:2305.06023.

Teaching

2024-2025 Organization UGent Algebra Seminar.

LECTURES

2025-current MT101SC: Differential Calculus (Maynooth University)
 2025-current MT341P: Rings and Fields 1 (Maynooth University)
 2025-current MT112A: Linear Algebra 1 (A) (Maynooth University)
 2023-2025 Linear Algebra and Geometry I (Ba 1 Mathematics and Physics, UGent)
 2023-2025 Algebra II: Fields and Galois theory (Ba 3 Mathematics, UGent)
 2020-2023 Introduction to Group Theory (Ba 1 Mathematics, VUB)

TUTORIAL SESSIONS

2023 Introduction to GAP at CIMPA crossroads of geometry, representation theory and higher structures
 2022-2023 Basic skills in Mathematics (Ba 1 of all Sciences, VUB)
 2022-2023 Mathematical Statistics (Ba 3 Mathematics, VUB)
 2017-2021 Ring and Module Theory (Ba 2 Mathematics, VUB)
 2017-2020 Probability Theory (Ba 2 Mathematics, VUB)
 2016-2023 Linear Algebra (Ba 1 Biology, VUB)
 2016-2017 Discrete Mathematics (Ba 1 Mathematics, VUB)
 2016-2017 Calculus (Ba 1 Biology, VUB)

Mentoring

PHD

2021-2025 S. Trappeniers - The combinatorics of set-theoretic solutions to the Yang-Baxter equation and related algebraic structures - FWO PhD Fellow (2021-2025), joint with L. Vendramin.

MASTER

2025 L. De Baecke - Guidance for the course “Seminar” at UGent
 2023-2024 M. Moreel - Links between set-theoretic solutions of the Yang-Baxter equation and algebraic structures (for the course “Literature study” at UGent)
 2020-2021 S. Trappeniers - Skew braces and the Yang-Baxter equation: an exposition on structure theorems (thesis)
 2017-2018 C. Verwimp - Braces and the Yang-Baxter equation (thesis) (joint with E. Jespers)

BACHELOR

2025 L. Meddens - Algebraic structures in knot theory (thesis)
 2025 Z. Van Eetvelde - Skew braces of prime power order (thesis)
 2021 B. D'Hauwer - Unique product groups and the unit conjecture (thesis)
 2021 S. Croux - Embedding semigroups in groups (thesis)
 2020 S. Frycia - To be nil or not to be nil: the extraordinary counterexample of A. Smoktunowicz (thesis)
 2017 (joint with E. Jespers) Jente Bosmans - Quandles and their automorphisms (thesis)

SELECTION OF TALKS

- 2026 Finiteness conditions on skew braces and solutions of the Yang-Baxter equation, Interplay of skew braces, Hopf-Galois structures and solutions of the YBE, University of Leeds, Leeds, UK
- 2025 Cabling of non-involutive solutions of the Yang-Baxter Equation, The Yang-Baxter equation and all that, Banach Center Bedlewo, Poland
- 2024 Groups, Rings and the YBE, Lund Algebra Seminar, University of Lund, Lund, Sweden
- 2024 Cabling non-involutive solutions of YBE, Yang-Baxter algebras, Heriot-Watt, Edinburgh, UK
- 2024 Finiteness conditions in Yang-Baxter algebras, Skew braces, Braids and the Yang-Baxter equation, Banff international research station, Canada
- 2023 Centrally nilpotent skew braces, Groups, rings and the Yang-Baxter equation, Blankenberge, Belgium
- 2023 The structure monoid and algebra of a left non-degenerate solution of the Yang-Baxter equation, Algebra seminar, Exeter, UK
- 2023 Nilpotency of skew braces: an overview, Oberwolfach mini-workshop: skew braces and the Yang-Baxter equation, Oberwolfach, Germany
- 2022 The structure algebra of set-theoretic solutions to the Yang-Baxter equation, Groups and Algebras in Bicocca 2022, Milan, Italy
- 2022 Skew braces, the Yang-Baxter equation and non-associative algebras, Workshop on non-associative algebras 2022, Covilha, Portugal
- 2022 Skew braces and the Yang-Baxter equation, Algebra, Geometry and Physics seminar, joint MPIM and HU Berlin, Berlin, Germany
- 2022 Central nilpotency of skew braces, Algebra Seminar Edinburgh University, Edinburgh, UK
- 2019 Factorizations of skew left braces, Oberwolfach mini-workshop: Algebraic tools to solve YBE, Oberwolfach, Germany
- 2018 Left semi-braces and solutions of the Yang-Baxter equation, Noncommutative and Non-associative structures, braces and applications, Malta
- 2017 Coleman automorphisms of finite groups and their minimal normal subgroups, AGTA: Advances in Group Theory and Applications, Lecce, Italy

Research stays

- 2026 Banach Center Bedlewo, Poland, Collaborative workshop between 26/01 and 31/01
- 2024 Leeds, UK, Visiting I. Colazzo between 18/11 and 22/11
- 2024 Warsaw, Poland, Visiting J. Okninski between 7/04 and 13/04
- 2023 St Andrews, UK, Visiting O. Konovalov between 15/05 and 20/05
- 2023 Exeter, UK, Visiting I. Colazzo and N. Byott between 10/04 and 15/04
- 2022 Edinburgh, UK, Visiting A. Smoktunowicz between 15th of November and 22nd of November
- 2022 Berlin, Germany, Visiting P. Tamaroff between 7th of November and 12th of November
- 2022 Warsaw, Poland, Visiting L. Kubat between 04/04 and 14/04
- 2019 Buenos Aires, Argentina, visiting L. Vendramin between 23rd of September and 8th of October
- 2019 Shanghai, China, visiting L. Vendramin, between 5th April and 17th April
- 2019 Lecce, Italy, visiting F. Catino, I. Colazzo and P. Stefanelli between 4th of February and 1st of March

Organized conferences

- 2023 local organizer: Doctor honoris causa seminar for Prof. Efim Zelmanov, VUB
- 2023 main organizer: Groups, rings and the Yang-Baxter equation, Blankenberge, Belgium
- 2019 local organizer: Groups, rings and associated structures, Spa, Belgium

2017 local organizer: Groups, rings and the Yang-Baxter equation, Spa, Belgium

Visitors

2026 R.Cascella, collaboration on finiteness properties of skew braces and solutions of the YBE

Evaluator selection committee

2023 For PhD-mandate (6 years) at VUB

2020 For professor position “Algebra: groups, rings or the Yang-Baxter equation” at VUB

2019 For professor position “Discrete techniques in data science” at VUB

Referee for international journals

Algebras and Representation Theory

Bulletin of the London Mathematical Society

Canadian Mathematical Bulletin

Communications in Algebra

International Mathematics Research Notices

Journal of Algebra

Journal of Algebra and its Applications

Journal of Combinatorial theory, series A

Journal of London Mathematical Society

Journal of Pure and Applied Algebra

Proceedings of the Edinburgh Mathematical Society

Proceedings of the Edinburgh Royal Society

Publicacions Matemàtiques

Evaluator PhD committee

2025 Jari Desmet, Non-associative Aspects of Algebraic Groups, UGent.

2022 Charlotte Verwimp, Set-theoretic solutions of the Yang-Baxter equation and associated algebraic structures, VUB.

Administrative tasks

AT UNIVERSITEIT GENT

2024-current Representative in department council

AT VRIJE UNIVERSITEIT BRUSSEL

2017-2023 Representative in department council

2017-2023 Representative in education council for mathematics (secretary: 2019-2023)

2019-2023 Representative in faculty education council

2020-2023 Representative in faculty council

2021-2023 Representative in faculty board

2020-2023 Representative in education quality council

Outreach to general public

2018-current	Participate with popularizing lecture in science show “Wauw!Wetenschap” in Herentals, Belgium.
2018-current	National jury member in ”Competition on Scientific expositions” for youth between 15-18 years.
2014-current	Member of the board of ”Natuur en Wetenschap VZW” a science-popularizing organization for youth in Belgium.

Language proficiency

Dutch, native speaker
English, fluent (C1 certificate)
French, intermediate
Italian, intermediate